

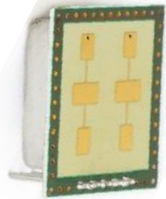
- **Classical RADARs are based on pulse-echo-technique**
 - distance determination by Time-of-Flight (ToF) measurement
 - see <https://de.wikipedia.org/wiki/Radar> => „Impulsradar“ => animation
- **Disadvantages of pulse-echo-RADARs**
 - long radio range requires **high power** (short) Tx-**pulses**
 - high range resolution requires **very short-time pulses**
- **One solution: FMCW-RADAR**

Tobias Otto, „Principles of FMCW Radars“, TU Delft,

see <http://de.slideshare.net/tobiasotto/principle-of-fmcw-radars>.

- slides 1-7: principle of FMCW radar
- slide 15: block diagram of an FMCW radar
- slide 12: FMCW radar signal processing

FMCW-Radar: how it works



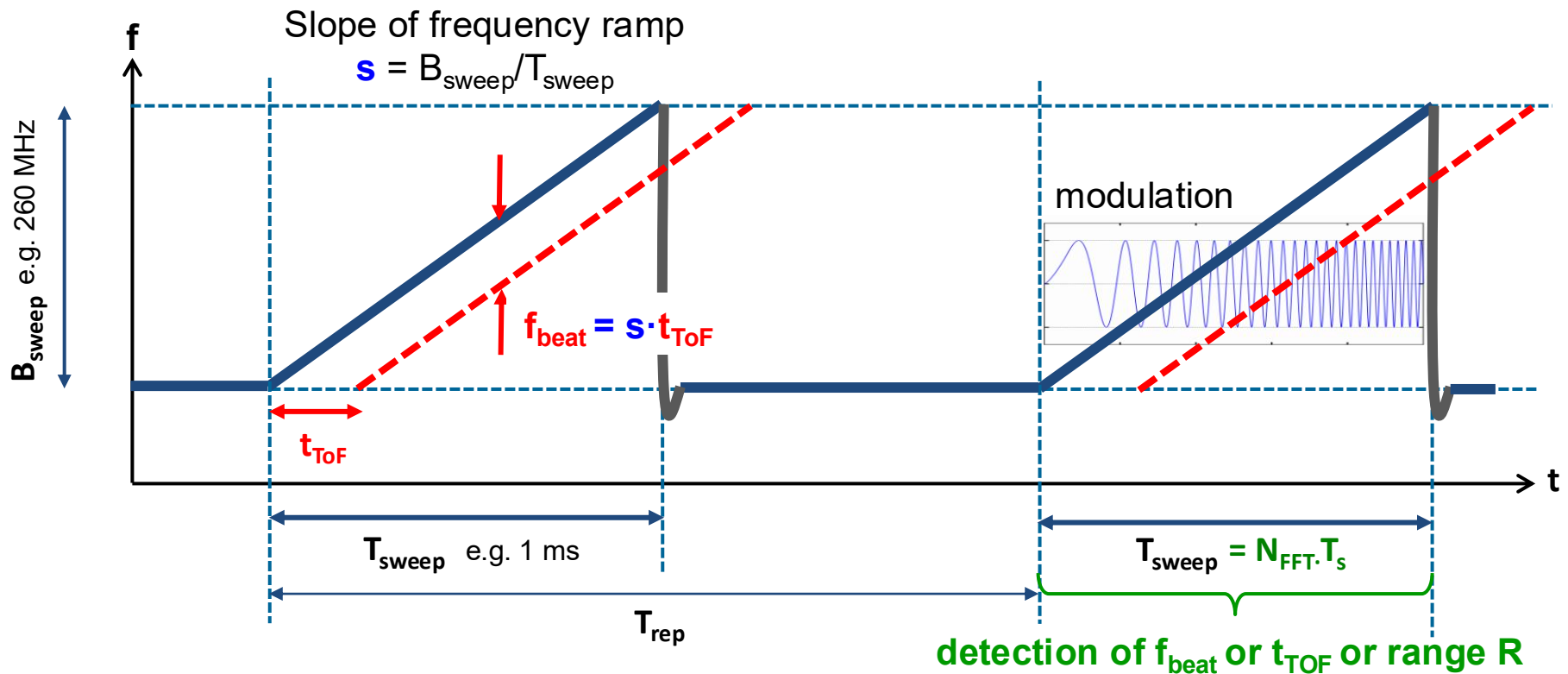
range $R=?$



Time-of-Flight
 $t_{\text{ToF}} = 2 \cdot R/c$



dynamic or
static target



- **FMCW-Radar: frequency-difference $f_{\text{beat}} \sim \text{range } R$**
 - Exercise: find $f_{\text{beat}} = \dots\dots\dots$
 - CW-Radar: frequency-difference $f_d \sim \text{Doppler-velocity}$
- **Rx-signal is multiplied with Tx-signal inside sensor**
 - frequency-difference appears at the sensor-outputs
e.g: $I(t) = A \cdot \cos(2\pi \cdot f_{\text{beat}} \cdot t - \varphi_0)$
 - detection of frequency difference f_{beat} with **FFT** of length N_{FFT}
- **Range-resolution $\Delta R = c / (2 \cdot B_{\text{sweep}})$**
 - e.g: $B_{\text{sweep}} = 260 \text{ MHz} \Rightarrow \Delta R = 0.6 \text{ m}$
 - Independent of ramp slope S and FFT-length N_{FFT}

FMCW-Radar Parameters

$c = 299\,792\,458\text{ m/s}$ (speed of light)

$B_{\text{sweep}} = 260\text{ MHz}$ (B_{sweep} is typically given)

$T_{\text{sweep}} = 1\text{ ms}$ (choice)

Slope $s = B_{\text{sweep}} / T_{\text{sweep}} = 260\text{ MHz} / 1\text{ ms}$ (indirect choice by T_{sweep})

$f_s = 64\text{ kHz}$ (choice)

Range $R = \frac{f_{\text{beat}} \cdot c \cdot T_{\text{sweep}}}{2 B_{\text{sweep}}} = \frac{f_{\text{beat}} \cdot c}{2 s}$ « / 2 » since it is an echo and the measurement path has been travelled twice.

Frequency resolution $\Delta f_{\text{beat}} = \frac{f_s}{N_{\text{FFT}}} = \frac{1}{T_{\text{sweep}}} = 1\text{ kHz}$ (= FFT resolution if $T_s \cdot N_{\text{FFT}} = T_{\text{sweep}}$)

Range resolution: $\Delta R = \frac{\Delta f_{\text{beat}} \cdot c \cdot T_{\text{sweep}}}{2 \cdot B_{\text{sweep}}} = \frac{c}{2 B_{\text{sweep}}} = 0.58\text{ m}$

Max. Range: $R_{\text{max}} = \frac{f_{\text{beat max}} \cdot c}{2 s} = \frac{\frac{f_s}{2} \cdot c}{2 s} = \frac{f_s \cdot c}{4 s} = \frac{f_s \cdot c \cdot T_{\text{sweep}}}{4 B_{\text{sweep}}} = 18.4\text{ m}$

→ R_{max} can be adjusted by adjusting T_{sweep} or f_s

If $T_s \cdot N_{\text{FFT}} = T_{\text{sweep}} \rightarrow R_{\text{max}} = \frac{f_s \cdot c \cdot T_s \cdot N_{\text{FFT}}}{4 B_{\text{sweep}}} = \frac{c \cdot N_{\text{FFT}}}{4 B_{\text{sweep}}}$

- **DAC generates sawtooth-signal for VCO-input**
 - Anti-Imaging-filter: RC-TP 1. order ($f_g = 10 \text{ kHz}$) is sufficient
 - VCO-Input voltage range: 0 bis 3.3 V
 - VCO-Input impedance 20 k Ω
- Sampling of sensor signal $I(t)$ (and maybe $Q(t)$)
 - Support of f_s up to 100 kHz
 - Bandwidth of analog Gain-Amplifier up to 50 kHz
Gain-Bandwidth-Product of PM3-OpAmp is 2 MHz
=> 2 amplifier stages are required!
- start of f-ramp and FFT-block must be «synchronized»